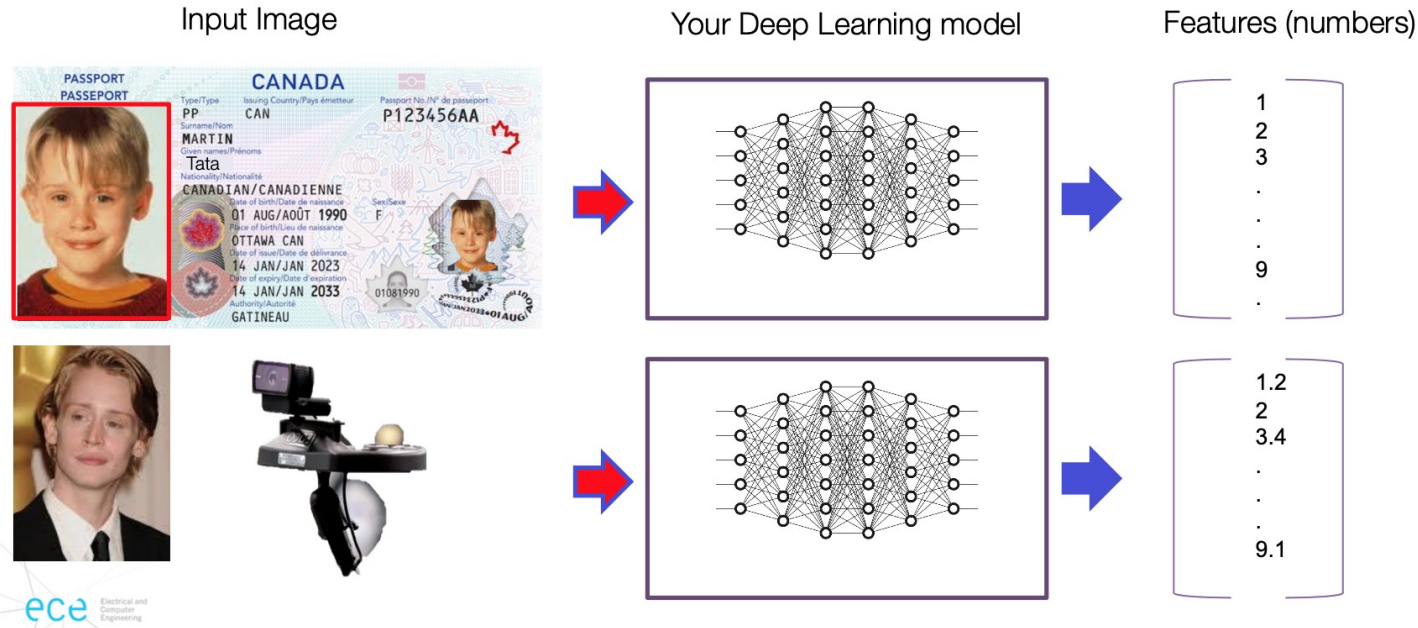


Age-Invariant Face Recognition Using Deep Learning

By: Eric Cao, Yanjun Meng, Akhil Rajavaram

Introduction - Goal

- Design a deep learning network to verify identity regardless of age



Dataset - M1SM (Training)

- Over 10 million images of nearly 100,000 individuals.
- Includes a wide range of ages, poses, and expressions.
- Large-scale dataset with wide age diversity.
- Original dataset contains noise and requires cleaning.



Dataset - B3FD (Training and Testing)

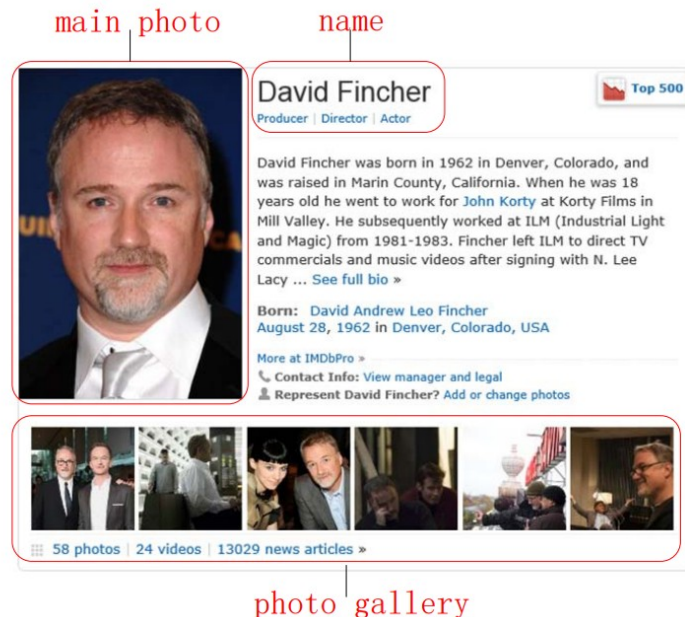
- 375,592 facial image samples with age labels and 53,759 unique individuals.
- Cleaned M1SM dataset of faulty samples for refined age estimation.
- Automated cleaning may introduce biases in age distribution or other characteristics.

Dataset	Images	DEX [41]			Mean-Variance [36]			DLDL-v2 [17]		
		R-MAE	A-MAE	ϵ -error	R-MAE	A-MAE	ϵ -error	R-MAE	A-MAE	ϵ -error
Appa-Real [1]	7,591	7.511	6.382	0.478	7.462	6.355	0.477	7.369	6.099	0.457
AgeDB [31]	16,488	10.023	9.303	0.536	10.378	9.625	0.548	9.851	9.251	0.549
MegaAge [47]	41,941	10.666	9.685	0.643	9.941	8.938	0.623	10.021	9.073	0.633
MORPH [40]	55,134	13.608	12.171	0.652	13.434	11.768	0.642	12.871	11.211	0.614
CACD [7]	163,446 ¹	13.538	12.454	0.637	12.555	11.409	0.613	12.079	11.005	0.605
IMDB-WIKI [41]	523,051 ²	7.155	6.673	0.433	7.077	6.259	0.431	6.568	6.101	0.418
IMDB-WIKI + CACD [41, 7]	686,497 ³	7.658	7.174	0.447	7.606	6.831	0.449	7.106	6.704	0.417
B3FD-CS [ours]	130,388	10.914	9.816	0.583	11.033	9.992	0.587	10.844	9.869	0.590
B3FD-IWS [ours]	245,204	5.340	5.072	0.388	5.394	4.808	0.386	5.158	4.684	0.383
B3FD [ours]	375,592	5.282	5.118	0.393	5.441	5.025	0.403	5.077	5.064	0.408

Comparison of B3FD to other datasets ([src](#))

Dataset - Casia-WebFace (Testing)

- 494,414 images of 10,575 subjects.
- Images scraped from the IMDB website semi-automatically.
- Used extensively for evaluating face recognition algorithms on standard benchmark
- Known for incorrect labels and images that do not belong to the labeled subject



“main photo” is used as initial seed and the 58 photos in the “photo gallery” need to be annotated ([src](#))

Dataset - AgeDB (Testing)

- 16,488 images of 568 distinct subjects.
- Manual Collection which ensures accurate, noise-free age labels.
- Smaller size in comparison to other datasets
- Limited number of subjects, better for benchmarking than training.



ID: Van Damme, Jean-Claude
Age: 27



ID: Douglas, Michael
Age: 35



ID: Dalton, Timothy
Age: 48



ID: Sinatra, Frank
Age: 56

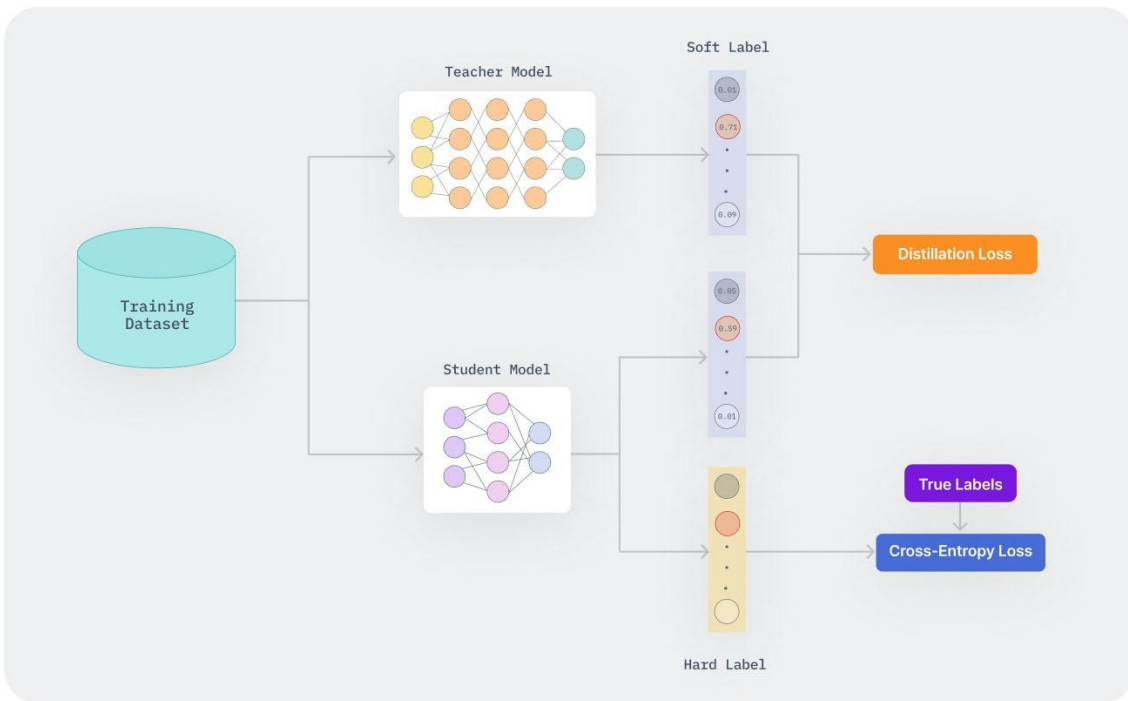


ID: Disney, Walt
Age: 64

Sample images from the dataset ([src](#))

Model - Knowledge Distillation

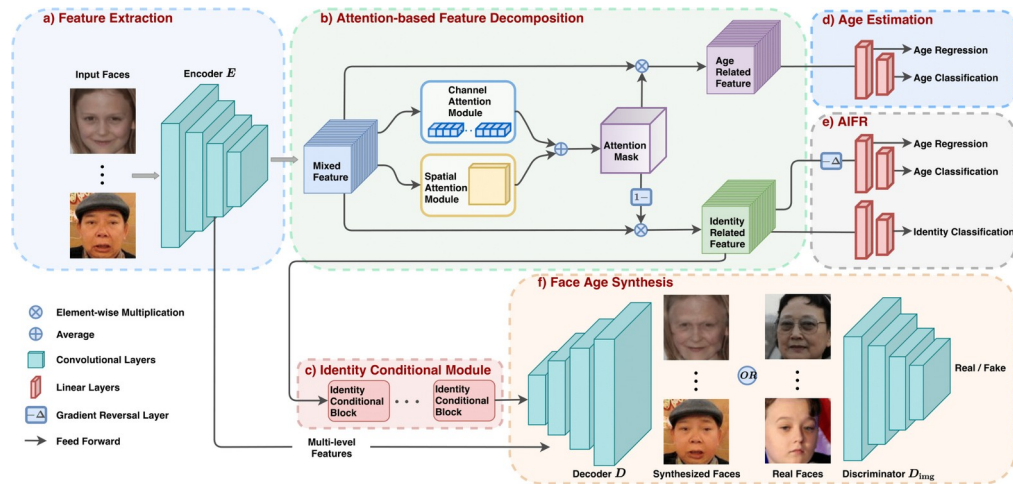
- Training a smaller student model to emulate a larger teacher model to distill knowledge.
- Reduces model size and computational requirements.
- Ideally student model learns to extract features that are invariant to age-related changes in the face.



Overview of the knowledge distillation training framework ([src](#))

Model - MTLFace

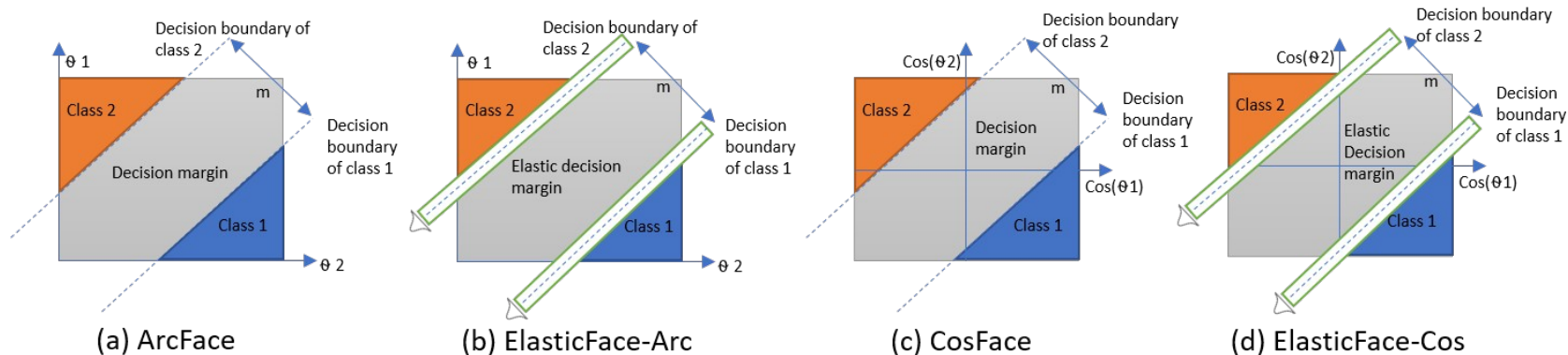
- Separates mixed face features into identity- and age-related components using an attention mechanism.
- Decorrelates identity features from age features, which is essential for AIFR as it reduces the intra-class discrepancy caused by aging.



Overview of the MTLFace methodology ([src](#))

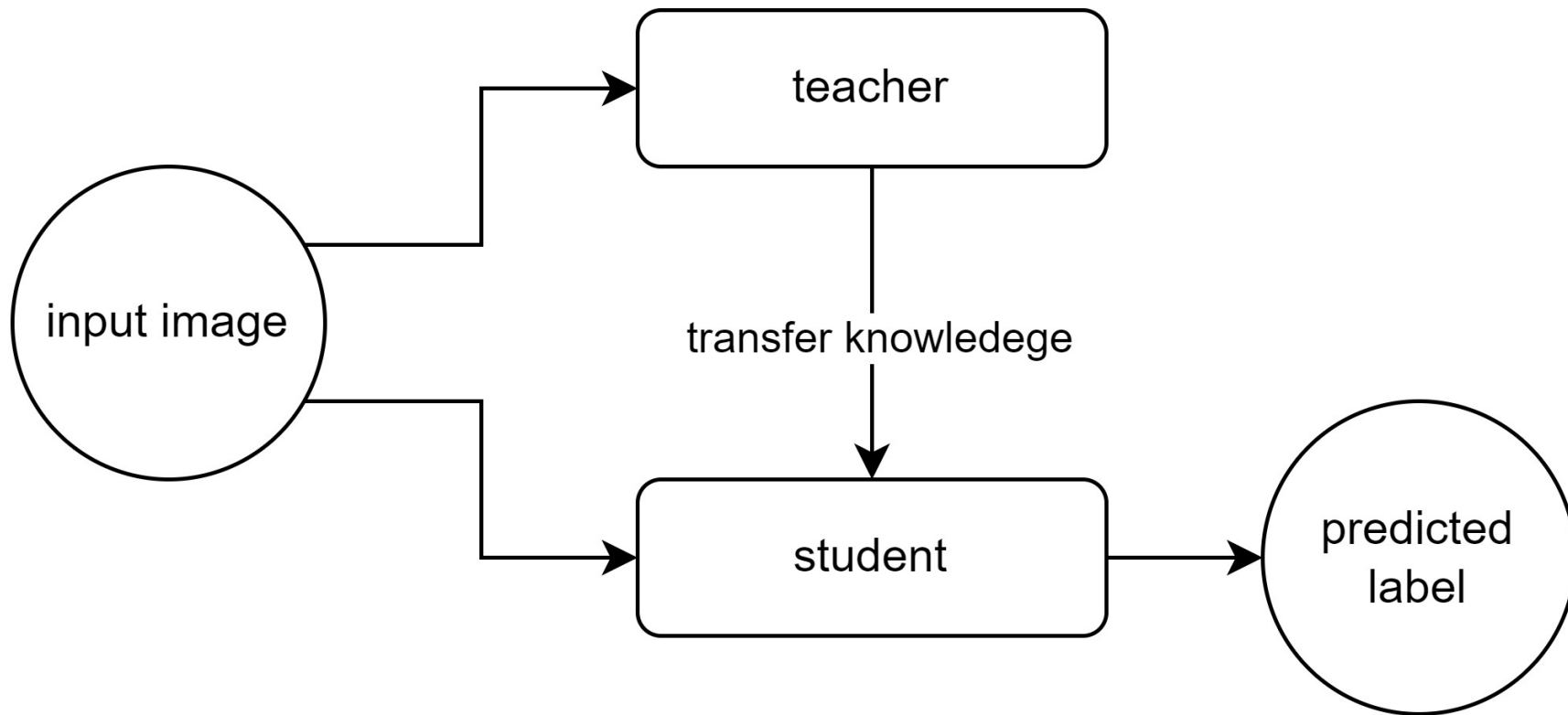
Model - ElasticFace

- The Elastic Margin Loss function introduces a dynamic margin into the training process.
- Employs a margin that is randomly drawn from a normal distribution during each training iteration.
- Allows the decision boundary to adjust dynamically to make features of the same individual more compact while ensuring features of different individuals are well-separated.
- Has the ability to learn more complex decision boundaries and better generalize to new data.



Depiction of how the dynamic margin affects decision boundaries ([src](#))

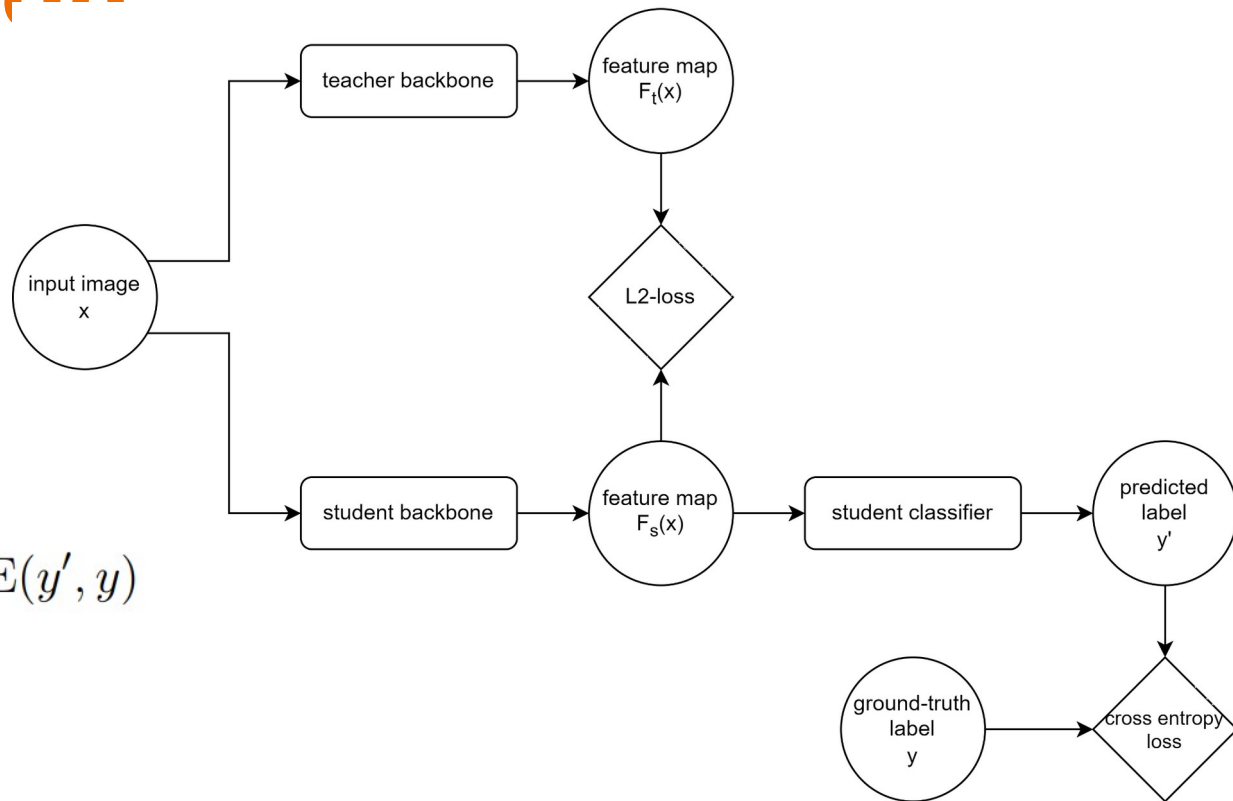
Methodology - OursKD



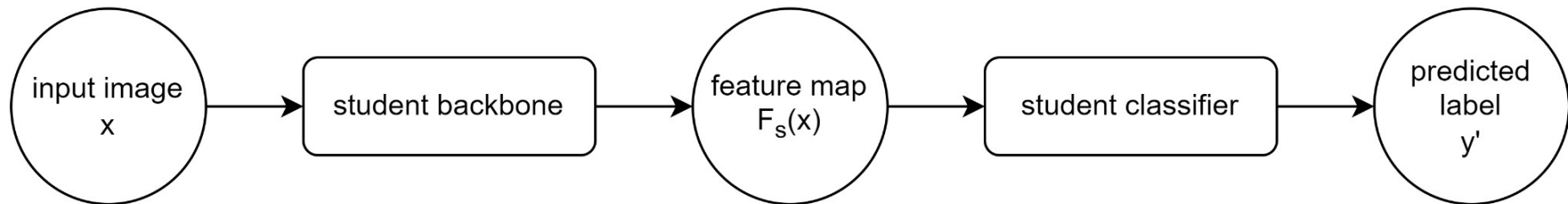
Loss function

- L2 loss
- Cross entropy loss

$$L = L2(F_t(x), F_s(x)) + CE(y', y)$$

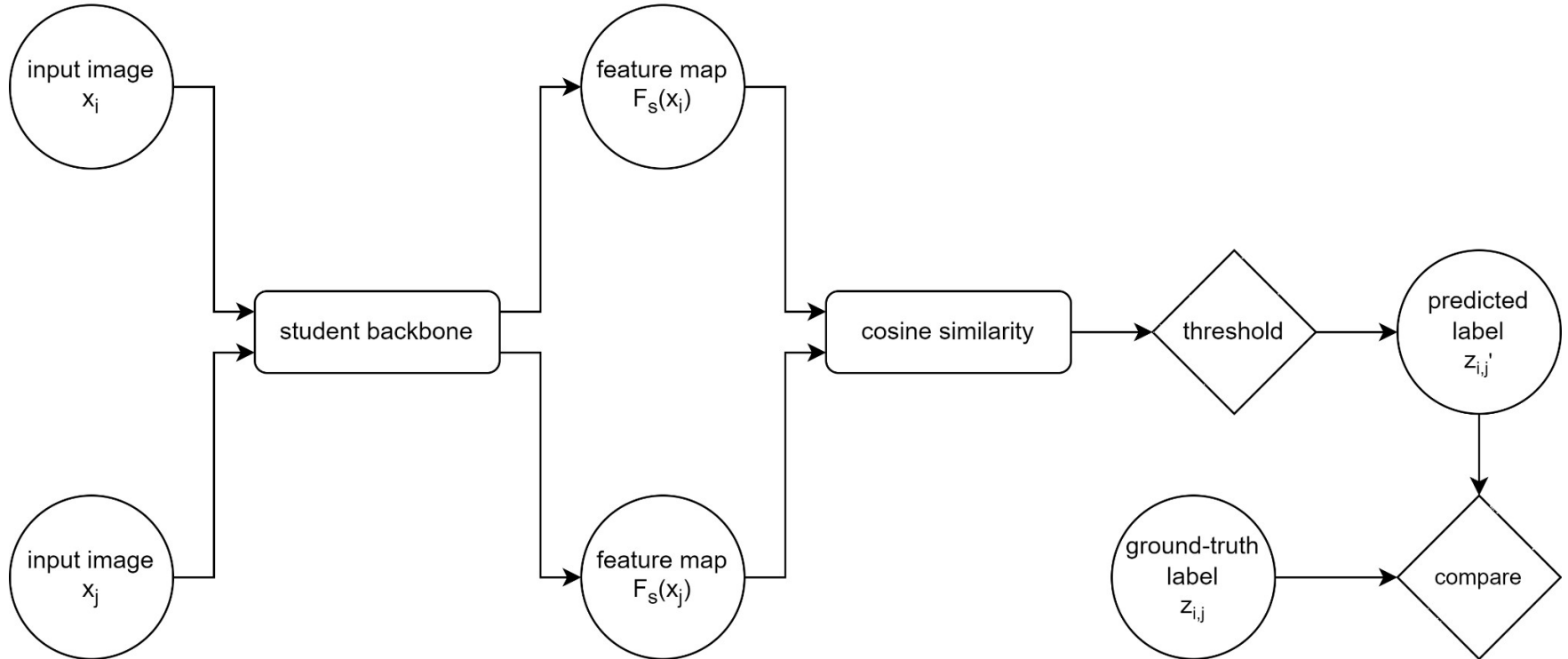


Performance Metrics



- Accuracy of prediction
- Representativeness of feature map

Performance Metrics



Results

Verification Accuracies of Individual Models

Model	Train set	Test accuracy		
		B3FD	AgeDB	CASIA
ElasticFace	B3FD	91%	57%	68%
MTLFace	B3FD	96.3%	77.4%	82.2%
ElasticFace	MS1MV2	87.1%	98.3%	96.1%

- First two rows trained by us from scratch on B3FD, evaluated on unseen datasets AgeDB and CASIA by us
- Last row trained by authors on MS1M, evaluated on new dataset B3FD by us

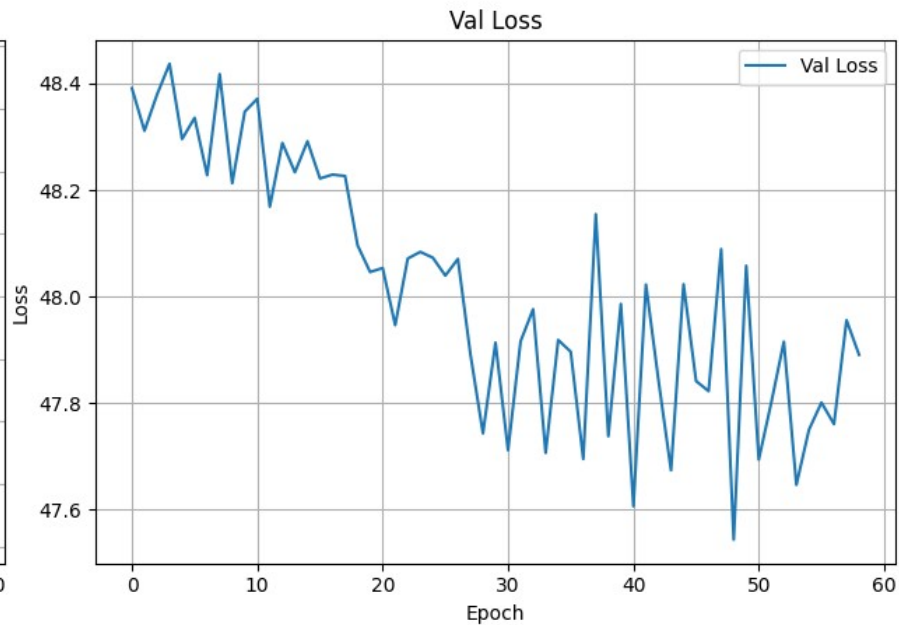
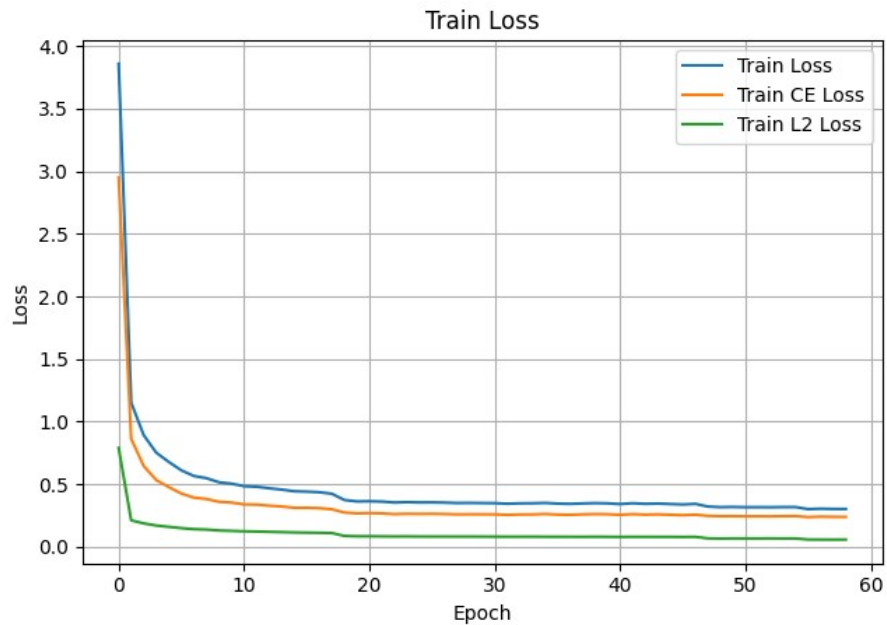
Verification Accuracies of KD Models

Model		Test accuracy		
Student-B3FD	Teacher-MS1M	B3FD	AgeDB	CASIA
ElasticFace	ElasticFace	95.7%	76.5%	82.1%
MTLFace	ElasticFace	96.0%	76.5%	82.6%

- Both student models were pretrained on B3FD, then trained alongside a teacher pretrained on MS1M, lastly evaluated on unseen datasets AgeDB and CASIA

Results

OursKD Learning Curve (student MTLFace, teacher ElasticFace)



Results

- Using (1) pretrained weights or architectures that differed from teacher (2) models that are pretrained on larger datasets helped improve verification accuracy and generalization capacity
- Reason: more diverse feature maps

(1) Accuracy with v.s. without using pretrained weights

Model		Test accuracy		
Student	Teacher-MS1M	B3FD	AgeDB	CASIA
MTLFace- scratch	ElasticFace	83.4%	59.6%	67.3%
MTLFace- B3FD	ElasticFace	96.0%	76.5%	82.6%

(2) Accuracy using B3FD- v.s. MS1M-trained teacher

Model		Test accuracy		
Student-B3FD	Teacher	B3FD	AgeDB	CASIA
ElasticFace	ElasticFace- B3FD	90.8%	64.7%	71.8%
ElasticFace	ElasticFace- MS1M	95.7%	76.5%	82.1%

Limitations

- Did not optimize training script for ElasticFace on B3FD → could have led to better baseline → altered subsequent comparisons
- Did not fine tune loss terms for using MTLFace as a student → suboptimal or conflicting effects → could not outperform individual

MTLFace model

Comparison between individual v.s. KD-ElasticFace model

Model		Test accuracy		
Student	Teacher	B3FD	AgeDB	CASIA
ElasticFace	n/a	91%	57%	68%
ElasticFace	ElasticFace	95.7%	76.5%	82.1%

Comparison between individual v.s. KD-MTLFace model

Model		Test accuracy		
Student	Teacher	B3FD	AgeDB	CASIA
MTLFace	n/a	96.3%	77.4%	82.2%
MTLFace	ElasticFace	95.7%	76.5%	82.1%

Future Work

- To transfer learned features more effectively
 - Fine tuning hyperparameters and other settings
 - Innovative loss terms
- To improve generalization
 - Impacts of dataset quality and scope
 - More new, unseen data
 - Stronger prediction head